

Project Proposal (Proposed Solution)

SMARTBRIDGE / SKILL WALL

Date: 29 September 2026

Team ID: SWTID-2026-3580

Project Title: EduGenie: Google Gemini Powered Learning Assistant

Maximum Marks: 5 Marks

Project Proposal (Proposed Solution) Report

EduGenie is a Google Gemini powered learning assistant designed to help students understand topics, ask questions, practice with quizzes, summarize long passages, and receive structured learning guidance.

Project Overview

Objective	Provide an AI-assisted learning workflow that gives clear explanations, answers questions, generates quizzes, summarizes study material, and re
Scope	Support common student learning tasks through a web interface connected to a FastAPI backend, Google Gemini services, and a lightweight loca

Problem Statement

Description	Students often face complex concepts, scattered learning resources, unanswered doubts, lengthy passages, and limited structured practice.
Impact	These difficulties can make learning slower and less structured. EduGenie provides a single workflow for explanation, Q&A, practice, summaries,

Proposed Solution

Approach	Use Google Gemini for Q&A, quiz generation, summarization, and learning-path recommendations, with LaMini-Flan-T5-783M supporting concept
Key Features	Q&A; concept explanation; 3-question MCQ quiz; long-passage summarization; beginner-to-advanced learning recommendations.

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Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Computing Resources	Local development/testing on available CPU/GPU hardware; lightweight local model
Memory	RAM	Sufficient memory for FastAPI, Gemini client, and local LaMini model; local execution
Storage	Application and model files	Project source, Python environment, model cache, and generated logs/results.

Software

Frameworks	Python frameworks	FastAPI, Uvicorn, Jinja2
Libraries / APIs	AI and supporting libraries	Google GenAI SDK, Transformers, PyTorch, python-dotenv, HTTPX
Development Environment	IDE / source control	VS Code or equivalent IDE; Git and GitHub

AI / Data

AI Services / Models	Model source and purpose	Google Gemini API for Q&A, quizzes, summaries and learning paths; LaMini-Flan-T
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